

Question ID: 34aca984

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

The length of a rectangle is **50** inches and the width is x inches. The perimeter is at most **210** inches. Which inequality represents this situation?

Answer

- A. $2x + 100 \leq 210$
- B. $2x + 100 \geq 210$
- C. $2x + 50 \leq 210$
- D. $2x + 50 \geq 210$

Correct Answer: A

Rationale

Choice A is correct. The perimeter of a rectangle is equal to the sum of **2** times its length and **2** times its width. It's given that the rectangle's length is **50** inches and the width is x inches. Therefore, the perimeter, in inches, is $2(50) + 2x$, or $100 + 2x$, which is equivalent to $2x + 100$. It's given that the perimeter is at most **210** inches; therefore, $2x + 100 \leq 210$ represents this situation.

Choice B is incorrect. This inequality represents a situation where the perimeter is at least, rather than at most, **210** inches.

Choice C is incorrect. This inequality represents a situation where **2** times the length, rather than the length, is **50** inches.

Choice D is incorrect. This inequality represents a situation where **2** times the length, rather than the length, is **50** inches, and the perimeter is at least, rather than at most, **210** inches.